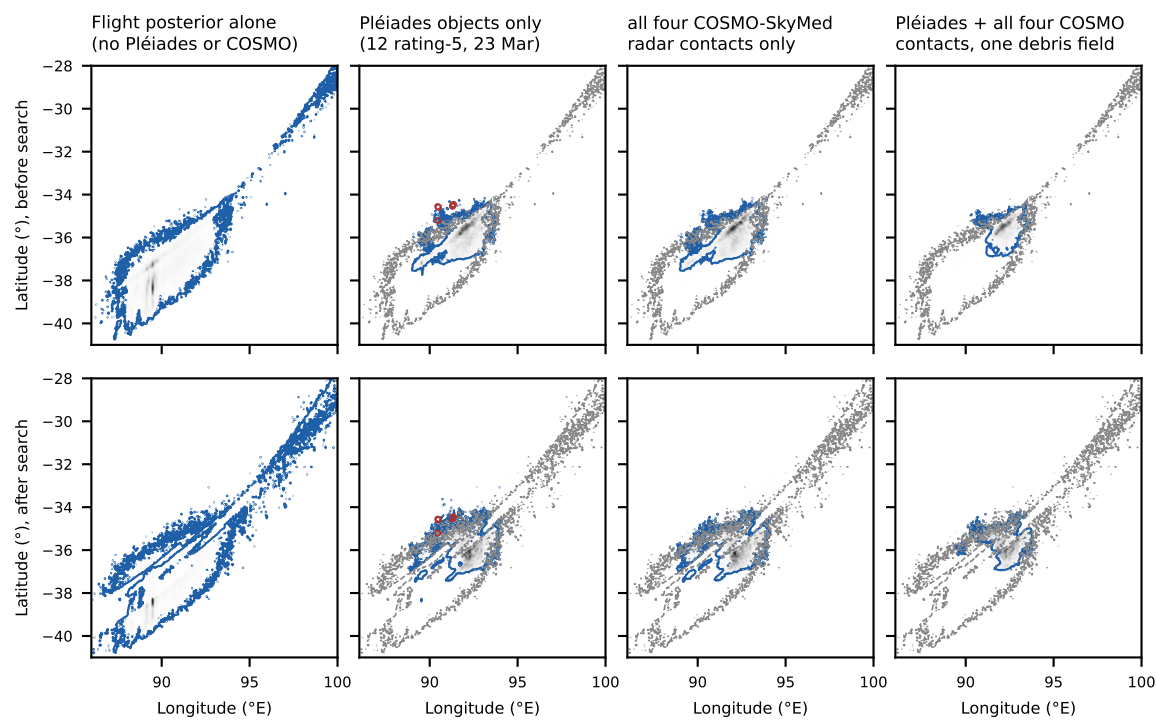


If the imaged objects are from 9M-MRO: where the debris entered the sea, before (top) and after (bottom) the Phase 2 + Bluefin-21 search — 00:19 R600 BTO + Raw BFO, +unpowered



Source: impacts next-free (4 seeds pooled with equal weight; source run src-next-free, code 3c6319f); prior track 289.7 deg (sd 1.0); config chain (run.json): davey2016-inmarsat + no-exhaustion-prior + s1-factor + s2-temperature + s3-internal + s4-ceiling + s5-hard-reject + reference-snapshots + radar-full + free + server-free + hypotheses/end-of-flight/smoke/snapshot-m0011.toml + hypotheses/end-of-flight/smoke/terminal.toml + hypotheses/end-of-flight/full/reference-289.toml + hypotheses/end-of-flight/full/descent-idle-floor.toml + hypotheses/end-of-flight/full/seed-1.toml; tempered sampler (m1839, m1941, m2041, m2141, m2241, m0011, 16 stages); particles per mode [500000, 250000, 1250000, 250000, 1250000]; BFO bias (run.json bfo_bias): {"mean_hz":150.0,"sd_hz":25.0}.

Labels: core (b); request-17 sampler fix IN, split-half NOT converged; two-tank bookkeeping only; internal-v1 fuel (one-engine flow 2x its tables); Inmarsat ephemeris; descent idle floor ON; EoF sweep run by an architecture stand-in; existence constraint variant (b) 'unpowered' (sensitivity); PROVISIONAL-OVERNIGHT; stratum next-free.

What was run: 00:19 R600 BTO + Raw BFO; R600 BTO (18,400 μ s) + R600 BFO (182 Hz) scored; R1200 not scored; BFO at face value (raw); log-on not from fuel exhaustion (no lag term) [end-of-flight arm: r600/no-offset]; log-on cause 'other' (no fuel-exhaustion log-on density). Weight = impact weight x exp(loglik:r600/no-offset) x end of flight's existence constraint 'unpowered' (its own constraint_log_factor; 'alive' = airborne at 00:19:37.443, PROVISIONAL-OVERNIGHT, 10 Oct ~04:05 UTC).

End-of-flight physics PROVISIONAL: dive class (b) (being rebuilt against Boeing's simulator runs); Boeing-calibrated glide (Pete's reference).

Search (row 2): the searched-areas module's own per-impact likelihood, base case Phase 2 2014-17 + Bluefin-21, point target, undetectable probability $\rho = 0.05$; Ocean Infinity 2018 not applied.

Under H (columns 2-4): ocean models GLORYS12 + ERA5 and GlobCurrent daily + ERA5, equal weight (P+C formed per model, then averaged); measured transport error (GDP-replay OU fit per component, no added diffusivity); windage 0-5 % uniform; released 00:20 UTC 8 Mar.

Pleiades: rating-5 objects merged into 6 clusters at 3 km, equal cluster weights ($\rho_{04} = 0$). COSMO-SkyMed: all four contacts, equal weights, dawn-20 Mar and dusk-21 Mar passes equally weighted; source, time and footprint unverified.

Shading: density normalised to its maximum. Solid blue: 90 % HDR of the panel's PDF. Dashed grey: 90 % HDR of the flight posterior at the same stage. Circles: Pleiades rating-5 clusters; triangles: COSMO-SkyMed contacts. Grid 0.05 deg; 0.3-0.9 % of impact weight lies outside 85-103 E, 43-25 S and is not shown.

Every PDF in columns 2-4 is conditional on H (the objects are debris from 9M-MRO); no Bayes factor or $P(H | \text{data})$ is computed or implied.

Label: next-free; 00:19 option r600/no-offset+unpowered; search Phase 2 + Bluefin-21; core (b): request-17 sampler fix IN, split-half NOT converged; two-tank bookkeeping only; internal-v1 fuel (one-engine flow 2x its tables); Inmarsat ephemeris; descent idle floor ON; EoF sweep run by an architecture stand-in; existence constraint variant (b) 'unpowered' (sensitivity); PROVISIONAL-OVERNIGHT; stratum next-free; generated by prepare/branch_eof289.py + branch_figure.py (hypothesis/pleiades).